

Previously Worked on Trial Project

Section 1 – Reading and Understand the Task:

Dataset:

- Given sample dataset with 8 columns, 4371 entries
 - 2021 to 2025 seasons
 - 2021-2024 is training set
 - 3498 entries
 - 2025 is prediction/testing/out of sample validation set.
 - 873 entries

Columns and Feature Decisions:

Column	Thoughts
Season	Not sure if I should include this in my model. I don't think that this provides much if I choose it as a feature. Only will be used in the code to separate the training and validation sets
Name	Providing identification information to the model would allow the model to make connections to names. I want the model to learn from the metrics exclusively so when there is an entry that was not in the training set, the model can predict based off the metrics.
Playerid	Same issue as name, I will not use this as a feature.
Team	Same issue as name and Playerid. Adding team would subtract from the models ability to understand what should be the most important metric, stuff+. There would be biases introduced that I want to avoid.
K%	Target variable
TBF	I will use this as a feature because it will allow my model to learn from the sample size of an entry. This will give the model a weighing ability to help the accuracy of predictions.

Stuff+	This should be the most important feature. There should be the largest correlation between K% and stuff+.
Age	I will use this as a feature because it does not have the same identification effect name, playerid or team do. It should not introduce a bias that needs to be avoided. I will do a test trial to see if it benefits or not.

Additional information and my thought process:

- I could research and include more features if I want.
 - This would take time.
 - Also, I cannot use that data for the validation season
 - Would it even make the model more accurate if I have to test with features missing?
 - My gut says that it would not help my model if it trains on data that it won't have when I test on the validation set.

Goal of the project:

- Predict K% for the out of sample validation set which is 2025 pitchers
- Submission is judged based on clarity of description of my methodology and how I present my findings
 - Secondary judge category is accuracy

How will I achieve this:

- I will track my notes and thoughts as I work clearly on this document.
- I will use python and machine learning modeling to achieve the lowest RMSE or other accuracy metric.
- Track RMSE over time to keep mental clarity for what works and does not work
- Clearly cite in my code and graphics when I used an LLM.

Section 2 - Modeling Information:

Model Selection:

- Which model will achieve the best results?
 - XGBoost
 - Handles complex, non-linear relationships well
 - Logistic regression will keep results between 0-1 which is what k% lives in
 - Use cross fold validation on XGBoost to assist finding the best hyperparameters
 - Use optuna to tune the hyperparameters
 - Experience working with XGBoost in previous projects is an added bonus
 - GAM or weighted GLM
 - Lower overfitting risk
 - Easier interpretability
 - A feature of this model is that it does not need assistance finding optimal hyperparameters. No need for optuna or cross fold validation
 - Lighter model, less code needed, faster to run
 - Ensemble
 - Blend the two models together to see if they can learn from their mistakes.

Accuracy Metric Selection:

- I think RMSE would be good because it will heavily penalize large errors. I want to be as accurate as possible.
- Mean absolute error would be a good option as well but does not penalize large errors like RMSE would.

Additional Model Thoughts:

- I do not believe I need to do any further feature selection work
 - Of the 8 features, there are 3 that are going to be included as features in my model that I know will be valuable.
 - The columns including playerid, name, and team information will introduce a bias to my model that I am unsure that I want.

- A more in-depth model that can understand the complex relationship between team specific pitching philosophy could potentially be useful to a model. Or even statcast's park factors, but this would assume that a pitcher always pitches at home.
- For this project, this would introduce bias that I want to ignore.
- A player's name or identifier introduces overfitting/memorization. I want my model to learn from the metrics and not which players are good at what. Giving identification information would make my model bias towards those and not the metrics. I want to avoid this.
- One way to check how accurate my model is to compare it to natural variance in pitcher K% season to season.
 - If my model is significantly higher than the average natural variance, then I will have concern.
 - This will help give me context to the accuracy of my model.
 - I will find this by sorting the players by their Playerid and their seasons.
 - I will calculate the year over year change of their K% and display them as RMSE and MAE equivalent numbers.
 - This will establish a baseline to compare my model's results to.

Section 3 - Trials:

MODEL 1 – XGBoost Regressor:

- Model information
 - XGBoost Regressor with reg:logistic
 - Optuna for hyperparameter tuning
 - GroupKFold for cross validation
 - K=4
- Terminal Output:

Best is trial 87 with value: 0.046946695774625166.

Best Parameters: {'n_estimators': 760, 'max_depth': 3, 'learning_rate': 0.012606072082010419, 'subsample': 0.5866295158149826, 'colsample_bytree':

```
0.6772479325042357, 'min_child_weight': 9, 'reg_alpha':  
0.005733781313188211, 'reg_lambda': 0.015689804304572515}  
Validation RMSE: 0.0604  
Validation MAE: 0.0454
```

- Analysis:
 - The MAE is saying that the model is incorrect by an average of 4.5%
 - RMSE being about 1.33x larger means there is not large outliers weighing down my model. Overall it is consistently around 4.5% error for each prediction.

MODEL 2 - GAM:

- Model Information
 - GAM
 - After testing a few times, I noticed the model was pinning to the edge of the grid and was getting worse on validation as the grid widened.
 - With the help of claude.ai, we determined that the GAM internal GCV/UBRE was not working and changed it to search lambda directly against the validation RMSE/MAE since that was the metric that I have chosen
 - This improved the results of the GAM Model.
- Terminal Output:

```
GAM Validation RMSE: 0.0617  
GAM Validation MAE: 0.0458
```
- Analysis:
 - Much faster and lighter model than XGBoost using optuna to find hyperparamters.
 - Very similar results for being a “lighter” model.
 - If I had restrictions that made using XGBoost difficult, such as an extremely large dataset with a less powerful machine, using the GAM would provide a benefit knowing that it can perform close to the XGBoost model.

MODEL 3 - Ensemble:

- Model Information
 - Blending GAM and XGBoost with ensemble technique.
- Terminal Outputs

XGBoost RMSE: 0.0604 || MAE: 0.0454

GAM RMSE: 0.0634 || MAE: 0.0462

Blend results (weight = share given to XGBoost):

xgb_weight=0.0	RMSE=0.0634	MAE=0.0462
xgb_weight=0.1	RMSE=0.0629	MAE=0.0461
xgb_weight=0.2	RMSE=0.0624	MAE=0.0459
xgb_weight=0.3	RMSE=0.0619	MAE=0.0458
xgb_weight=0.4	RMSE=0.0615	MAE=0.0457
xgb_weight=0.5	RMSE=0.0612	MAE=0.0456
xgb_weight=0.6	RMSE=0.0609	MAE=0.0455
xgb_weight=0.7	RMSE=0.0607	MAE=0.0455
xgb_weight=0.8	RMSE=0.0606	MAE=0.0454
xgb_weight=0.9	RMSE=0.0605	MAE=0.0454
xgb_weight=1.0	RMSE=0.0604	MAE=0.0454

Best blend: xgb_weight=1.0 RMSE=0.0604 MAE=0.0454
- Analysis:
 - Clearly XGBoost outperforms GAM. The ensemble method gave full weight to the XGBoost model.

Pitcher's Natural Variance:

- Baseline used to evaluate my models' results
- Terminal Output:

Number of year-over-year pairs: 2442

Year-over-year K% change stats:

Mean absolute change (MAE-equivalent): 0.0515

RMSE-equivalent (sqrt mean squared change): 0.0760

Std dev of change: 0.0756

Median absolute change: 0.0368

Year-over-year change by TBF bucket:

tbf_bucket	mean	count
<50	0.084205	426
50-150	0.056974	507
150-300	0.045734	851
300+	0.033621	658

Section 4 - Final Analysis:

Comparing Model Results:

Baseline		Model 1 - XGBoost		Model 2 – GAM		Model 3 – Ensemble	
MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
5.15%	7.60%	4.54%	6.04%	4.58%	6.17%	4.54%	6.04%

- The XGBoost is my best performing model.
 - Ensemble has the same accuracy metrics because the best version of the ensemble model was giving full weight to the XGBoost model and zero weight to the GAM.

Thoughts:

- After trying 3 different versions of my strikeout percentage models, I have come to the conclusion that my XGBoost performs the best.
 - Final RMSE was 0.0604 and Final MAE was 0.0454
 - This means the model was off by an average of 4.54% percent on the predicted strikeout percentages.
 - The RMSE being about 1.33 times my MAE means that there were not many outliers dragging my model down. My model was overall good at predicting the whole validation set regardless of outlier sample size pitchers or outlier stuff+ numbers.
 - I am satisfied by that result. Knowing that my model is consistent is a good thing. Perhaps I could be more accurate overall, especially if I include more features, but

having a consistent model is something that I am proud of when looking at my results.

- When I compare my model to the natural variance of year-to-year strikeout percentages, my model outperforms that number.
 - This is significant because it tells me that my model is successful in identifying trends from the metrics that were provided. For a 3-feature model, I believe this is a very good result.